

- | | |
|-------------|-------------|
| P1: (3, 7) | P2: (4, 6) |
| P3: (5, 5) | P4: (6, 4) |
| P5: (7, 3) | P6: (6, 2) |
| P7: (7, 2) | P8: (8, 4) |
| P9: (3, 3) | P10: (2, 6) |
| P11: (3, 5) | P12: (2, 4) |

- $$\text{Distance}(A(x_1, y_1), B(x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

[illegible]

DBSCAN Clustering Algorithm Solved Example – 1

minPts = 4 and epsilon (ϵ) = 1.9

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1: (3, 7)	0											
P2: (4, 6)	1.41	0										
P3: (5, 5)	2.83	1.41										
P4: (6, 4)	4.24											
P5: (7, 3)	5.66											
P6: (6, 2)	5.83											
P7: (7, 2)	6.40											
P8: (8, 4)	5.83											
P9: (3, 3)	4.00											
P10: (2, 6)	1.41											
P11: (3, 5)	2.00											
P12: (2, 4)	3.16											

DBSCAN Clustering Algorithm Solved Example – 1

minPts = 4 and epsilon (ϵ) = 1.9

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
P1: (3, 7)	0											
P2: (4, 6)	1.41	0										
P3: (5, 5)	2.83	1.41	0									
P4: (6, 4)	4.24	2.83	1.41	0								
P5: (7, 3)	5.66	4.24	2.83	1.41	0							
P6: (6, 2)	5.83	4.47	3.16	2.00	1.41	0						
P7: (7, 2)	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8: (8, 4)	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9: (3, 3)	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10: (2, 6)	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
P11: (3, 5)	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12: (2, 4)	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

DBSCAN Clustering Algorithm Solved Example – 1

minPts = 4 and epsilon (ϵ) = 1.9

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
<u>P1</u>	0											
P2	<u>1.41</u>	0										
<u>P3</u>	2.83	<u>1.41</u>	0									
<u>P4</u>	4.24	2.83	<u>1.41</u>	0								
P5	5.66	4.24	2.83	1.41	0							
P6	5.83	4.47	3.16	2.00	1.41	0						
P7	6.40	5.00	3.61	2.24	1.00	1.00	0					
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0				
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0			
P10	<u>1.41</u>	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0		
<u>P11</u>	2.00	<u>1.41</u>	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0	
P12	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0

P1: P2, P10
P2: P1, P3, P11
P3: P2, P4

DBSCAN Clustering Algorithm Solved Example – 1

$\text{minPts} = 4$ and $\text{epsilon} (\epsilon) = 1.9$

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12	
P1	0												P1: P2, P10
P2	1.41	0											P2: P1, P3, P11
P3	2.83	1.41	0										P3: P2, P4
P4	4.24	2.83	1.41	0									P4: P3, P5
P5	5.66	4.24	2.83	1.41	0								P5: P4, P6, P7, P8
P6	5.83	4.47	3.16	2.00	1.41	0							P6: P5, P7
P7	6.40	5.00	3.61	2.24	1.00	1.00	0						P7: P5, P6
P8	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0					P8: P5
P9	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0				P9: P12
P10	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0			P10: P1, P11
P11	2.00	1.41	2.00	3.16	4.47	4.24	5.00	5.10	2.00	1.41	0		P11: P2, P10, P12
P12	3.16	2.83	3.16	4.00	5.10	4.47	5.39	6.00	1.41	2.00	1.41	0	P12: P9, P11

DBSCAN Clustering Algorithm Solved Example – 1

$\text{minPts} = 4$ and $\text{epsilon} (\epsilon) = 1.9$

Point	Status
P1	Noise
P2	Core
P3	Noise
P4	Noise
P5	Core
P6	Noise
P7	Noise
P8	Noise
P9	Noise
P10	Noise
P11	Core
P12	Noise

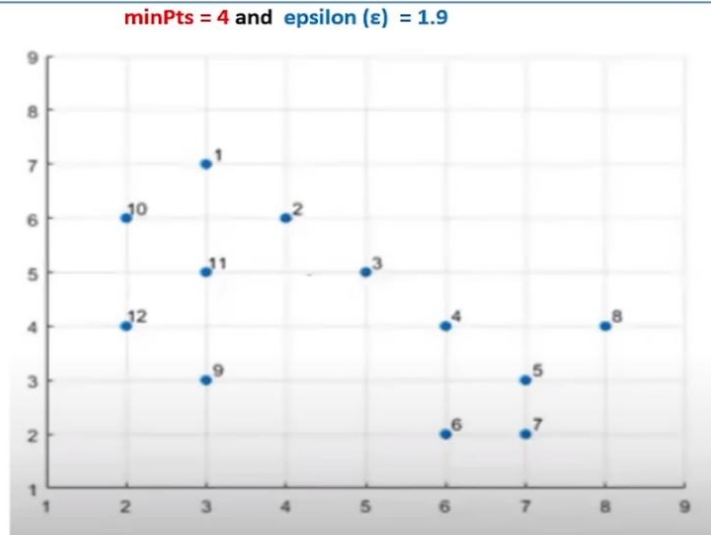
DBSCAN Clustering Algorithm Solved Example – 1

$\text{minPts} = 4$ and $\text{epsilon} (\epsilon) = 1.9$

Point	Status
P1	Noise
P2	Core
P3	Noise
P4	Noise
P5	Core
P6	Noise
P7	Noise
P8	Noise
P9	Noise
P10	Noise
P11	Core
P12	Noise

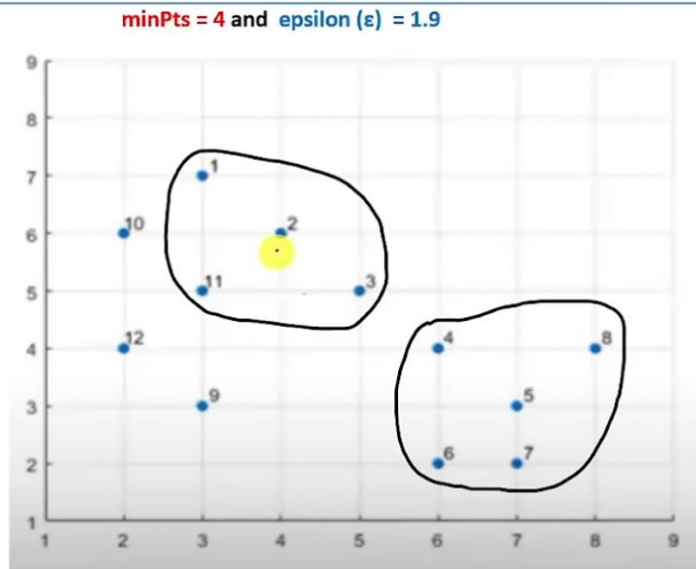
DBSCAN Clustering Algorithm Solved Example – 1

P1: P2, P10
P2: P1, P3, P11
P3: P2, P4
P4: P3, P5
P5: P4, P6, P7, P8
P6: P5, P7
P7: P5, P6
P8: P5
P9: P12
P10: P1, P11
P11: P2, P10, P12
P12: P9, P11



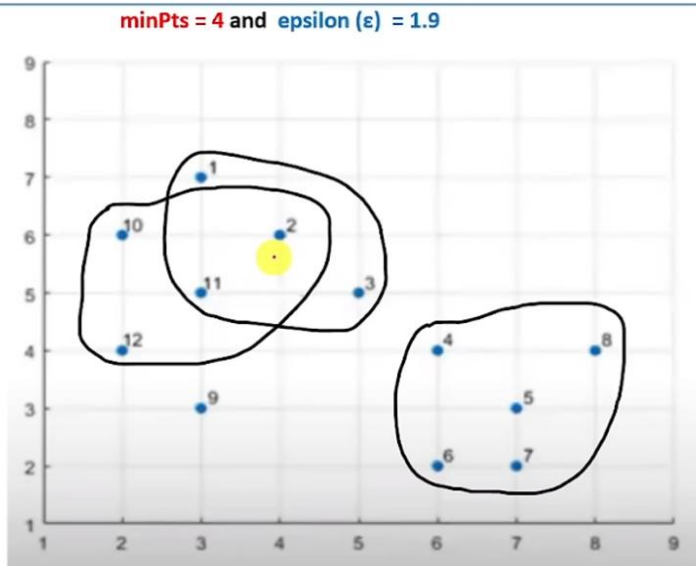
DBSCAN Clustering Algorithm Solved Example – 1

P1: P2, P10
P2: P1, P3, P11
P3: P2, P4
P4: P3, P5
P5: P4, P6, P7, P8
P6: P5, P7
P7: P5, P6
P8: P5
P9: P12
P10: P1, P11
P11: P2, P10, P12
P12: P9, P11



DBSCAN Clustering Algorithm Solved Example – 1

P1: P2, P10
P2: P1, P3, P11
P3: P2, P4
P4: P3, P5
P5: P4, P6, P7, P8
P6: P5, P7
P7: P5, P6
P8: P5
P9: P12
P10: P1, P11
P11: P2, P10, P12
P12: P9, P11



Spectral Clustering (in short):
Spectral clustering is a clustering technique that uses **graph theory and eigenvalues (spectrum) of a similarity matrix** to group data points. Instead of directly clustering raw data (like K-means), it first converts data into a graph and then clusters based on that structure.

◆ Key Idea

- Treat each data point as a **node in a graph**
 - Connect nodes using similarity (distance, Gaussian kernel, etc.)
 - Compute the **Laplacian matrix** of the graph
 - Use its **eigenvectors** to reduce dimensions
 - Apply a simple clustering algorithm (like K-means) on transformed data
-

◆ Steps

1. Create similarity matrix (W)
2. Compute degree matrix (D)
3. Compute Laplacian:

$$L = D - W$$

4. Find eigenvectors of (L)
 5. Use top eigenvectors → apply K-means
-

◆ Example

Suppose you have **two concentric circles of points**:

- Inner circle → Cluster 1
- Outer circle → Cluster 2

👉 Problem:

Algorithms like K-means fail because clusters are **non-linear**.

👉 Spectral Clustering Solution:

- Builds similarity graph (points close to each other are strongly connected)
 - Separates graph based on connectivity
 - Correctly identifies the two circles as separate clusters
-

◆ When to Use

- Non-convex clusters (complex shapes)

- Graph-based data (social networks, image segmentation)
-

◆ Simple Intuition

Instead of grouping by distance in space, it groups by **connectivity and relationships** between points.